



Assignment 5 Solution

✓ SECTION A — 2 Marks Questions (Answer in brief)

1. [CO5] Define Explanation-Based Learning (EBL).

Answer:

Explanation-Based Learning (EBL) is a form of **knowledge-based learning** where the system uses **prior domain knowledge** to generalize a single example into a broader concept. It explains why a specific example is an instance of a concept and then abstracts the explanation to form a general rule.

Example: Learning the general rule for “stack safety” after understanding one example of safe stack operation.

2. [CO5] What is a Markov Decision Process (MDP)?

Answer:

A **Markov Decision Process (MDP)** is a mathematical framework for modeling **decision-making in stochastic environments**.

It consists of:

- **S:** Set of states
- **A:** Set of actions
- **$P(s'|s,a)$:** Transition probability
- **$R(s,a)$:** Reward function
- **γ :** Discount factor

It satisfies the **Markov Property**, i.e., the next state depends only on the current state and action.

3. [CO5] State the difference between collaborative and content-based filtering.

Aspect	Content-Based Filtering	Collaborative Filtering
Basis	Uses user's past preferences and item features	Uses preferences of similar users

Aspect	Content-Based Filtering	Collaborative Filtering
Data Required	Item attributes	User-item interaction data
Example	Movie recommendation based on genre liked	Movie recommendation from users with similar taste

4. [CO4] What is Q-Learning?

Answer:

Q-Learning is a **model-free reinforcement learning** algorithm that learns the value of actions in states to maximize cumulative reward.

The Q-value update rule is:

$$Q(s, a) = Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

It does not require knowledge of the environment's transition probabilities.

5. [CO5] Define Temporal Difference (TD) Learning.

Answer:

Temporal Difference (TD) Learning is a reinforcement learning method that updates value estimates based on **difference between successive predictions** rather than waiting for the final outcome.

$$V(s) = V(s) + \alpha[r + \gamma V(s') - V(s)]$$

It combines ideas from Monte Carlo and dynamic programming.

✅ SECTION B — 7 Marks Questions (Short Essays)

1. [CO5] Explain the working of the FOCL algorithm with a suitable example.

Answer:

The **FOCL (First-Order Combined Learner)** algorithm extends **Explanation-Based Learning (EBL)** by integrating **Inductive Logic Programming (ILP)**.

It combines background knowledge and examples to learn first-order logical rules.

Steps:

1. **Input:** Background knowledge (BK), positive & negative examples.
2. **Construct Explanation:** Use BK to explain positive examples.
3. **Refine Hypothesis:** Use inductive bias to generalize from examples.
4. **Generate Rules:** Combine deductive and inductive inferences to form new hypotheses.

5. **Evaluate:** Test the generated rules on unseen data.

Example:

If BK includes facts about “bird $\rightarrow$ flies” and the system sees example “penguin does not fly”, FOCL generalizes the rule:

“IF bird AND NOT penguin THEN flies.”

Advantage: Combines the strengths of EBL (deduction) and ILP (induction).

2. [CO5] Describe the basic components of a Reinforcement Learning system.

Answer:

A **Reinforcement Learning (RL)** system has four main components:

1. **Agent:** Learner or decision-maker.
2. **Environment:** The external system the agent interacts with.
3. **Policy (π):** The agent’s behavior — mapping from states to actions.
4. **Reward Signal (R):** Feedback from the environment for each action.
5. **Value Function (V):** Expected long-term return from each state.
6. **Model (optional):** Predicts environment dynamics.

Working:

- Agent observes state $\rightarrow$ chooses action $\rightarrow$ environment responds with reward $\rightarrow$ agent updates policy.
Goal: Maximize cumulative long-term reward.
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3. [CO5] Compare content-based and collaborative filtering approaches in recommender systems.

Answer:

Feature	Content-Based Filtering	Collaborative Filtering
Data Type	Item attributes (genre, features, etc.)	User-item rating matrix
Dependency	Independent of other users	Depends on user similarity
New User Problem	Handles well if user profile exists	Suffers from cold start
New Item Problem	Struggles (no prior data)	Handles easily once rated
Accuracy	Personalized but limited diversity	Provides diverse recommendations

Feature	Content-Based Filtering	Collaborative Filtering
Example	Spotify recommends similar songs	Netflix recommends what similar users watch

✅ **Hybrid recommender systems** often combine both methods to overcome limitations.

✅ **SECTION C — 11 Marks Questions (Long Essays)**

1. [CO5] Discuss the Explanation-Based Learning (EBL) process with suitable diagrams and examples.

Answer:

Explanation-Based Learning (EBL) uses prior knowledge to derive generalizations from a single example by constructing an explanation.

Steps in EBL Process:

1. Training Example: Input a specific example of a concept.
2. Domain Theory: Knowledge base containing rules and facts.
3. Explanation: Derive why the example belongs to the concept.
4. Generalization: Abstract irrelevant details → form general rule.
5. Operationalization: Convert rule into an efficient form for future use.

Diagram:

Example + Domain Theory



Explanation



Generalized Rule



Future Application

Example:

Given example: "A stacked block structure is stable."

EBL uses knowledge about "base supports top" to generalize rule:

"If each block is supported, the whole stack is stable."

Advantages:

- Learns from few examples
- Produces human-readable rules

- Reduces search space

Limitations:

- Requires correct domain theory
- Sensitive to incomplete knowledge

2. [CO5] Explain in detail the Q-Learning algorithm, including its mathematical formulation and convergence properties.

Answer:

Q-Learning is a model-free, off-policy RL algorithm that learns optimal action-value functions.

Q-Function:

$Q(s, a)$ = Expected cumulative reward from taking action a in state s

Update Rule:

$$Q(s, a) = Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

where,

- α = learning rate
- γ = discount factor
- r = reward
- s' = next state

Algorithm Steps:

1. Initialize $Q(s, a)$ arbitrarily.
2. For each episode:
 - Choose action a using ϵ -greedy policy.
 - Observe reward r and next state s' .
 - Update $Q(s, a)$ using above rule.
3. Repeat until convergence.

Convergence Property:

If all state-action pairs are visited infinitely often and α decays appropriately, $Q(s, a) \rightarrow Q^*(s, a)$ (optimal Q-value).

Advantages:

- Works in unknown environments.
- Simple and effective for discrete state spaces.

Limitations:

- Slow convergence for large state-action spaces.
- Needs exploration–exploitation balance.

3. [CO4] Describe Markov Decision Processes (MDPs) and explain how they are used in reinforcement learning for decision making.

Answer:

Markov Decision Process (MDP) is the foundation of Reinforcement Learning (RL) that models sequential decision-making under uncertainty.

MDP Components:

1. S : Set of states
2. A : Set of actions
3. $P(s' | s, a)$: Transition probability
4. $R(s, a)$: Reward function
5. γ : Discount factor ($0 \leq \gamma \leq 1$)

Bellman Equation for Optimal Value Function:

$$V^*(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s' | s, a) V^*(s')]$$

Policy (π): Mapping from states to actions.

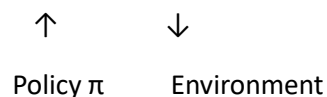
The goal is to find optimal policy π^* that maximizes cumulative rewards.

Usage in RL:

- RL uses MDPs to define agent–environment interaction.
- Q-learning, SARSA, and policy iteration are algorithms derived from MDP principles.

Diagram:

State (s) $\rightarrow$ Action (a) $\rightarrow$ Reward (r) $\rightarrow$ Next State (s')



Applications:

- Robotics path planning
- Game AI
- Automated control systems